

Practice Quiz 03C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 03

Negation of a quantified conjunction.

1. Consider the statement

$$\forall x \in \mathbf{R} \exists y \in \mathbf{R} (x < y \wedge y \leq x + 2).$$

- (a) Write its negation using no $\neg$ symbols, and name the logical laws you use.

Solution The quantifier negation laws change $\forall x \exists y$ to $\exists x \forall y$. De Morgan's law changes the conjunction to a disjunction of negations. Negating the inequalities then gives

$$\exists x \in \mathbf{R} \forall y \in \mathbf{R} (x \geq y \vee y > x + 2). \quad \blacksquare$$

- (b) Determine whether your negation is true or false, and justify your answer by choosing a suitable y for an arbitrary real x .

Solution The negation is false. For every real x , choose $y = x + 1$. Then $x < y$ and $y \leq x + 2$, so both $x \geq y$ and $y > x + 2$ are false. Thus no x makes the disjunction true for every y . $\blacksquare$